Automatic Pill Identification System based on Deep Learning and Image Preprocessing

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***Abstract*—The pharmaceutical industry annually distributes thousands of different prescription medications. However, the high volume of available medication increases the likelihood of errors during the dispensation process. These errors can result from distractions, incorrectly filed prescriptions, or similarities between pills. This paper proposes an automated system that employs computer vision to identify pills, aiming to enhance pill identification for pharmaceutical workers and consumers and reduce dispensing errors. The proposed approach involves creating a deep learning model using Keras, preprocessing pill images through an image data generator, and leveraging tools such as OpenCV and Paddle OCR to identify critical aspects of a pill, including its shape, color, and imprint. The ultimate goal is to develop a real-time pill identification system that can seamlessly integrate with video cameras, thereby facilitating smoother operations in high-volume medication dispensaries.**

Keywords—pill identification, deep learning, CNN, image preprocessing, feature extraction

I. INTRODUCTION

Throughout an individual's life, there are occasions when prescription medicines are necessary to address health issues. In the United States alone, there are approximately 6,800 prescription medications available, along with an extensive array of over-the-counter medications. However, the sheer number of available medications can give rise to certain challenges. Alarmingly, medication errors lead to fatal consequences for an estimated 7,000 to 9,000 individuals annually in the United States. A medication error refers to a preventable event that results in incorrect medication usage while the medication is under the control of a professional, patient, or consumer [1].

Medication errors can occur due to various factors. For instance, pharmacists may face distractions, time constraints, short-staffing, disorganization, and difficulties in differentiating between similar pills. To address these challenges, some pharmacies have implemented measures such as barcodes, carousel fill processes, and automated sorting systems to reduce dispensing error rates [2]. A study conducted in 2003 revealed that dispensing errors occur approximately four times a day for every 250 prescriptions filled. Furthermore, with an error rate of just 1%, an estimated 30 million errors would occur annually in the United States [3].

Pharmacists are tasked with manually identifying pills, and the tediousness of this process can contribute to errors, especially when combined with their numerous other responsibilities. Approximately 53% of a pharmacist's time is spent in the dispensary, while the remaining time is dedicated to management, counseling, and indirect services [4]. Automating the pill identification process can enhance efficiency, reduce fatigue, and alleviate the burden on pharmacists. OpenCV, a library for image analysis and computer vision, can be utilized to identify various aspects of a pill [5]. A convenient and viable tool for this purpose is a camera, particularly the ubiquitous phone camera, which opens up the possibility of consumer use [6]. With these considerations in mind, our study aims to design convolutional neural networks and implement image preprocessing methods to enable the automatic identification of pills through cameras or videos.

The contributions of this paper can be summarized as follows: (1) the construction of a pill identification dataset, (2) the training of two CNN models for pill detection and type classification, and (3) the development of an image preprocessing and database comparison method for pill identification using non-deep learning techniques. Initially, we created small datasets of pill images, which were utilized in conjunction with convolutional neural networks. This approach achieved a minimum accuracy of 99.8% in pill identification. However, when larger datasets were employed, it became apparent that the initial model was inadequate and prone to overfitting. To address this issue, a new model was created based on the first model but with fewer layers. Multiple aspects of a pill, including its shape, imprint, and color, were considered to improve accuracy. Additionally, we conducted a comparison between relying solely on the extraction of these features and utilizing CNNs. The resulting model achieved a training accuracy of 98.9% and a testing accuracy of 97.9%.

II. RELATED WORK

Previous research has primarily focused on the development of pill identification systems, along with studies exploring various techniques and methods that could contribute to the construction of such systems.

One study implemented an automatic segmentation approach, color identification, pill shape recognition, and optical

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recognition of imprints, combined with string matching, to identify pills [7]. Another investigation involved the creation of a mobile application that utilized the camera on a Windows mobile-equipped phone and a personal computer (PC) to estimate the size of pills using available homography for measurements [8]. In a different study, pill identification was achieved through feature extraction and matching with a database. Notably, the project utilized a limited dataset of 40 pill images, which were augmented to 36 images each [9]. Another approach utilized one of five CNN models specific to major shape types, utilizing the shape information to identify the correct pill [10]. Other research employed Multilayer Perceptron classifiers and Support Vector Machines for automated pill classification [11], feature extraction, preprocessing, and edge detection for pill classification [12], and compared k-means and Otsu's thresholding methods for imprint analysis, focusing on a particular aspect of pill identification [13]. Additionally, a Deep Convolutional Network (DCN) was developed to classify images of 400 pills taken in a laboratory setting [14]. Delgado conducted a study in which a deep-learning application for pill identification was created, followed by a comparison of various image classifiers based on different models [15].

Object recognition plays a vital role in this study, particularly considering the ultimate objective of utilizing a phone camera for automatic pill detection. One study compared RetinaNet, SSD, and YOLOv3 in terms of accuracy and detection speed for pill recognition. The results indicated that YOLOv3 achieved faster detection speed and proved suitable for real-time performance [16]. In another study, image preprocessing techniques were applied to optimize the training set for YOLOv3, which involved adjustments such as reducing the ratio between training and recognition images [17]. YOLOv3 has also been employed in other applications, including traffic counting and tomato harvesting, showcasing its versatility [18, 19].

Furthermore, other relevant studies explored various techniques and libraries applicable to machine learning and image analysis. For instance, one study investigated the utilization of ReLU instead of softmax as the classification layer and as an activation function at each layer of the neural network [20]. Another study provided an introduction to OpenCV, a library specifically designed for image and video analysis [5]. Lastly, for a more comprehensive understanding of real-time object recognition on phones, [6] delves into the subject in relation to CNNs.

III. METHODOLOGY

A. Dataset Preparation and Tools

The Pillbox dataset, created by the National Library of Medicine, was selected for its extensive collection of over 20,000 pill types. However, it should be noted that the pillbox was retired on January 28, 2021, and therefore, any pills manufactured after that date are not included in the dataset. The downloaded dataset consisted of only one image for each pill type, which was insufficient for our approach since we required a minimum of ten images for each pill in our initial sample set.

For the first sample set, we specifically chose a smaller subset consisting of two pills: Hydroxyzine and Prozac. These particular pills were selected because we had direct access to the actual medications in various dosages. This sample set served as the basis for our initial CNN implementation and the first iteration of the mobile application, which was developed using Android Studio.

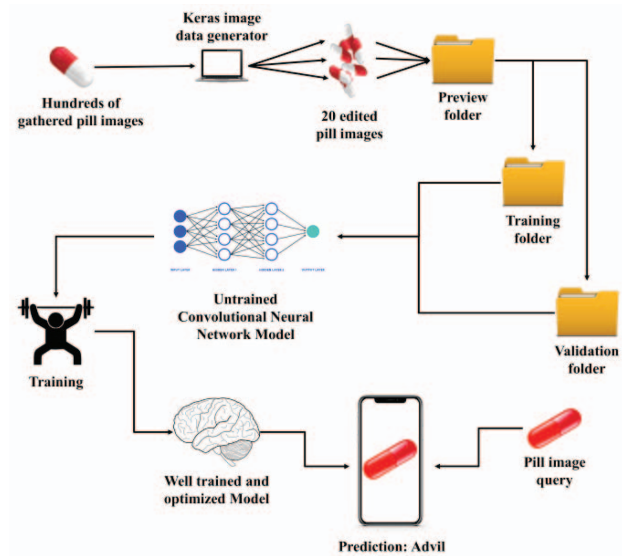


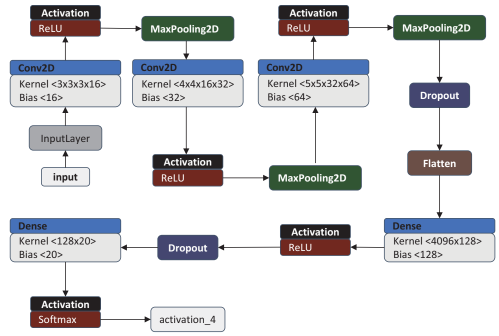
Fig. 1. System framework.

Furthermore, we created another sample set consisting of 20 pills, with 320 images per class in the training folder and 18 images in the validation folder. Figure 1 illustrates the system design for the model using this dataset. Additionally, we generated a larger sample set with 50 pills, aiming for at least 290 images per pill. However, due to the presence of discontinued pills in the dataset, it has yet to be tested. To address this, we are currently working on identifying replacement pills that are currently in production to revise the dataset.

After creating the image sample sets, we applied an image data generator to augment the images. This generator performs various transformations on the existing data, resulting in the creation of hundreds of images with different modifications. The transformations include changes in brightness, rotation, width and height shifting, ZCA whitening, rescaling, zooming, and horizontal flipping. The dataset consists of 20 pill varieties, with each variety fully represented by two pills, totaling 320 images

(a) Preimage data generator (b) Post-image data generator

Fig. 2. Sample image generated by the data generator.




```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 conv2d (Conv2D)             (None, 126, 126, 16)      448

 activation (Activation)     (None, 126, 126, 16)      0

 max_pooling2d (MaxPooling2D  (None, 63, 63, 16)       0
 )

 conv2d_1 (Conv2D)           (None, 60, 60, 32)        8224

 activation_1 (Activation)   (None, 60, 60, 32)        0

 max_pooling2d_1 (MaxPooling  (None, 30, 30, 32)       0
 2D)

 conv2d_2 (Conv2D)           (None, 26, 26, 64)        51264

 activation_2 (Activation)   (None, 26, 26, 64)        0

 max_pooling2d_2 (MaxPooling  (None, 8, 8, 64)         0
 2D)

 dropout (Dropout)           (None, 8, 8, 64)          0

 flatten (Flatten)           (None, 4096)              0

 dense (Dense)               (None, 100)               409700

 activation_3 (Activation)   (None, 100)               0

 dropout_1 (Dropout)         (None, 100)               0

 dense_1 (Dense)             (None, 20)                2020

 activation_4 (Activation)   (None, 20)                0

=================================================================
Total params: 471,656
Trainable params: 471,656
Non-trainable params: 0
_________________________________________________________________
```

Fig. 3. Model Diagram and Summary

per class. Consequently, the training dataset comprises approximately 6,400 images, while the validation and testing set each contains 360 images. Figure 2 displays a sample image generated through the data generator.

For the creation of deep learning models, we utilized the Keras API, which also facilitated the formulation of the image data generator. Keras, as an API for TensorFlow, offers ease of use in setting up convolutional neural networks and reduces the time required for experimentation. Additionally, we employed OpenCV for the image segmentation process, which involved isolating the pills from the background to perform color analysis. To scan and identify the text imprints on the pills, we utilized Paddle OCR. Lastly, if necessary, we used Pillow to open and edit images as a data type for pixel-level image manipulation.

B. CNN

In this section, we trained two CNN models: one for pill detection and the other for pill classification. The first model was designed for pill detection rather than identification. We utilized a sigmoid function at the end of the model, resulting in binary decisions of zero or one to indicate the presence of a pill in the image. The CNN would guess eleven out of the twenty classes in the twenty-class sample set at once if any trace of a pill was detected in an image. This model was also utilized in the initial iteration of the mobile application. To orient the model to a classification function, a second model was trained with ReLU and two softmax layers were used in conjunction. Softmax functions take a vector of real numbers as input and transform it into a probability distribution over multiple classes or categories. The softmax function calculates the exponential of each element in the input vector, normalizes the resulting values, and ensures they sum up to 1. The output would have decimals and the largest of these numbers would be the user's answer. When comparing two pills, the accuracy was high, and as the dataset was scaled higher with each new pill added, slight modifications were made to ensure the pills were still identified accurately. The latest model, as shown in Figure 3, currently employs only one softmax layer, as it demonstrated higher accuracy compared to using two softmax layers.

The aforementioned ReLU is a crucial activation function in the model. ReLU deactivates any neurons with a value less than zero, enabling the network to learn by generating new neurons that learn to recognize the characteristics of the image. When there is a fifty percent chance that a pill is a match, the activation function assigns a fifty percent probability to each pill. ReLU deactivates any neurons with a value less than zero, enabling the network to learn by generating new neurons that learn to recognize the characteristics of the image.

C. Mobile Application

The initial attempt to create a mobile application involved using the first sample set and the CNN model that employed the sigmoid function. The sample set consisted of approximately five pills, including Hydroxyzine and Prozac, with various dosage strengths. At the time, we had not considered using YOLOv3 for real-time object detection and instead utilized Keras and the TensorFlow Lite Model for real-time detection. Additionally, we tested the captured images on the Windows application to ensure accurate results in case the mobile application encountered any issues. Real-time detection using the TensorFlow Lite model yielded favorable outcomes. However, further examination led to the decision to deprecate the mobile application in favor of exploring the utilization of YOLOv3 for improved object detection and enhancements to the CNN model.

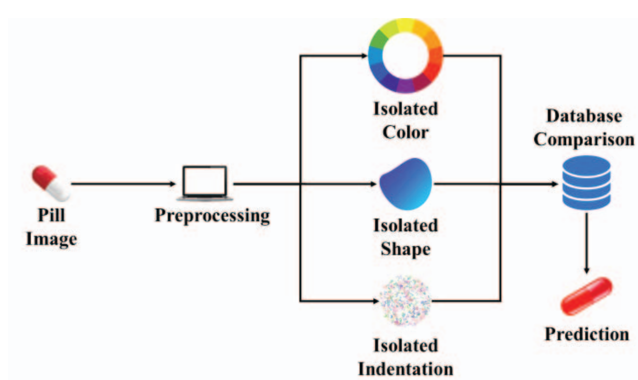


Fig. 4. Preprocessing pipeline.

(a) Original Image (b) Contoured Pill (c) Shape Extraction

Fig. 5. Shape Extraction.

D. Image Preprocessing

With the second CNN model currently in place, our focus shifted toward preprocessing the pills for identification. We proposed a second method for pill identification without CNN models. This method has two steps: (1) image feature extraction and (2) comparing with a database for pill identification. This step serves two purposes: to enhance pill identification accuracy and compare the results with the CNN methods without preprocessing. By performing feature extraction as a preliminary step, we aim to supplement the CNN model's identification capabilities. In cases where the CNN exhibits significant inaccuracies in one or two categories, alternative accurate data sources are relied upon. Figure 4 illustrates our plans for the preprocessing method for our system. To validate the effectiveness of feature extraction, we conducted tests using a single pill, cataloging all its parameters manually into the database for comparison. The integration of feature extraction with the CNN model proved instrumental in pill identification, primarily through shape extraction and segmentation, which will be further discussed in this study.

Regarding shape analysis, the system checks for the presence of large contours, indicating the creation of a distinct shape. Accurate color analysis relies on precise shape analysis. For imprint recognition, we employed Optical Character Recognition (OCR) techniques. An alternative approach involves thresholding the darkened parts of the pill, as indentations usually exhibit shadows in the images.

To facilitate segmentation, an HSV (hue, saturation, and value) layer is applied to extract the pill from the background. Unlike RGB, HSV provides a more accurate color scheme for manipulation and enables the determination of average color. By leveraging HSV, the average color can be easily calculated and compared, as each value represents a different metric of color rather than values within the color itself. Once the shape is identified, the average color is used to proceed with the process, followed by the utilization of OCR. Alternatively, the color information can be used to identify the shape.

1) Shape

First, the shape is extracted from the image using OpenCV2, an image editing library that enables image manipulation. OpenCV2 offers functions to threshold specific colors or alter colors based on predefined values. The segmentation process separates the pill, represented as a large shape in the foreground, from the background image. Figure 5 shows the before and after results of segmenting the foreground with the pill from the background, as well as the final extracted shape of the pill. The extracted shape is then classified into various categories, such as circle, ovoid, oval, or triangle, by comparing it against a shape database. If the segmentation fails to produce a proper separation, resulting in multiple small thresholds instead of two distinct parts, the process is considered inaccurate, and the analysis skips to utilizing OCR without proceeding to color analysis. By employing feature extraction with the CNN, the focus of the CNN is directed towards segmenting the foreground, while the background is handled during preprocessing.

2) Color

After identifying the contour and locating the pixels within it, the next factor to consider is color. Identifying the pill's contour allows us to isolate the pill's color within the contour region. However, it is worth noting that many pills share similar colors, causing some overlap and necessitating further processing. To find the contour, we remove the HSV value with the lowest brightness. In the HSV color space (hue, saturation, value), finding the contour based on brightness is achievable as the foreground pixels typically have higher brightness than the background pixels. Therefore, the segmentation process based on color analysis becomes crucial. Another approach is to focus on the center of the image, as pills captured in pictures are typically in the foreground. HSV can then be used to extract the image, while anything at the borders can be segmented into a separate layer. This alternative method allows shape identification through color rather than the reverse approach.

3) OCR

Paddle OCR is employed to extract the text from the pill image. The OCR algorithm relies on the imprints found on the pill surface, allowing the identification of a pill solely by analyzing the characters present, even if the previous two factors (shape and color) prove to be inaccurate. The pill is considered identified when the OCR successfully recognizes a set of words or numbers due to the fact that the majority, if not all, of pills, possess unique inscriptions. Figure 6 displays the results of the OCR when used. Additionally, the dosage information is typically included on the back of the pill, further narrowing down the possibilities in cases where the inscription is not unique. However, if a pill lacks any inscription, the identification process relies solely on color, shape, and the CNN model.

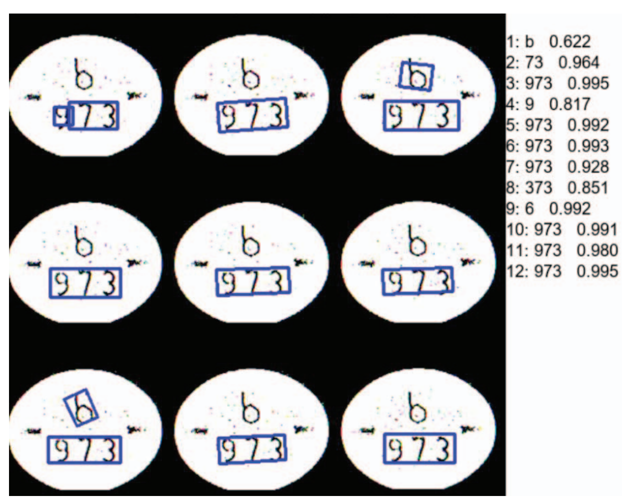


Fig. 6. Text Extraction.

IV. Results and Analysis

As mentioned earlier, the initial CNN model tested multiple classes simultaneously if any trace of a pill was detected in an image, due to the utilization of the sigmoid function. However, the mobile application utilizing this model encountered a limitation in its ability to detect pills under non-white light conditions. To address this issue, the sigmoid function was replaced with softmax and ReLU layers, and further development of the mobile application was discontinued in favor of overall neural network enhancements. Table 1 shows the individual accuracies of each pill class in the 20 pill dataset. The table compares the accuracies found utilizing only the CNN and the accuracies found when using the CNN in conjunction with the preprocessing method.

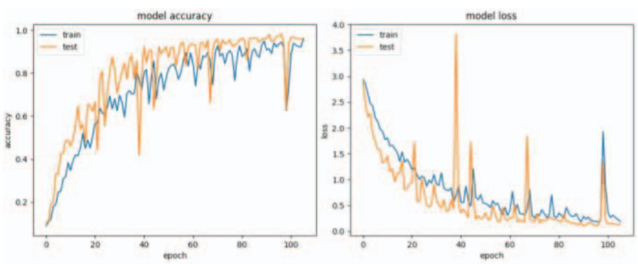


Fig. 7. Model accuracy and loss.

The feature extraction and analysis demonstrated a 97.9% accuracy rate when compared with the database. The 0.4 difference in accuracy between the results with and without the feature extraction can be attributed to the segmentation. If there is a threshold that is not larger than a certain size and near the same color as which is in the database, then the program disregards the segmentation step and moves on to the next step. This is intended as an error catching mechanism for the segmentation as the segmentation step does result in potentially unusable thresholds. It is important to note that this assessment was conducted on a smaller sample set consisting of a single type of pill, serving as a test for the functionality of the feature extraction process. This process involved manually adding parameters such as shape, color, and imprint of each pill to the database, along with other characteristics. The manual approach, while inconvenient, did have the benefit of slightly improved results, primarily due to more accurate segmentation when the CNN focused on segmenting the pill itself without being distracted by the background.

Currently, the model employing a single softmax layer achieves an accuracy of 98.9% during training, and the validation and test sets exhibit accuracies of 98.9% and 97.5%, respectively. These metrics pertain to the 20-pill sample set. The model accuracies and losses for training and test sets are shown in Figure 7.

V. Conclusion

The results of this study demonstrate the usefulness of a CNN in conjunction with feature extraction for pill identification. A convolutional neural network proves to be an effective tool in this context. However, it should be noted that updating the CNN with new pills added to the dataset is time-consuming. On the other hand, feature extraction and identification without the CNN require more processing power and time, but offer slightly higher accuracy. Additionally, preprocessing necessitates the use of a database, unlike CNN which does not.

Table 1. Accuracy for test batch of 20 sample pills for CNN.

Pill classes	Accuracy	Accuracy with preprocessing
Adderall 5mg	100%	100%
Adderall 7.5mg	88.8%	94.4%
Adderall 10mg	94.4%	94.4%
Adderall 12.5mg	100%	100%
Adderall 15mg	100%	100%
Adderall 20mg	72.2%	77.7%
Adderall 30mg	100%	100%
Hydroxyzine 25mg	94.4%	94.4%
Prozac 10mg	100%	100%
Synthroid 25mcg	100%	100%
Synthroid 50mcg	100%	100%
Synthroid 75mcg	100%	100%
Synthroid 88mcg	100%	100%
Synthroid 100mcg	100%	100%
Synthroid 112mcg	100%	100%
Synthroid 137mcg	100%	100%
Synthroid 150mcg	100%	100%
Synthroid 175mcg	100%	100%
Synthroid 200mcg	100%	100%
Synthroid 300mcg	94.4%	94.4%
Total Accuracy for Test:	97.5%	97.9%

The CNN approach presented in this study is particularly well-suited for high-volume dispensaries. It provides near-instantaneous results and requires minimal understanding of how pills are labeled. Compared to a system relying on a database and feature extraction, the CNN is significantly faster and requires less data storage. The difference in accuracy between the CNN (97.5%) and feature extraction (97.9%) is merely 0.4%. Although there are areas for improvement, such as scalability and training speed, the results thus far are promising.

Convolutional neural networks are remarkable tools that should be utilized in various disciplines due to their accuracy and speed. They can accurately predict pill identification, and their scalability makes them easily adaptable to a larger range of outputs. In contrast, manual identification based on pill traits would be slow and cumbersome. Future work could involve updating the dataset of fifty pills to exclude discontinued ones and include all currently circulating pills from the PillBox dataset. Prioritizing further work on the CNN, training a new model based on the expanded dataset, and acquiring real-world pills corresponding to the dataset for testing in a realistic environment are important steps. Furthermore, the implementation of YOLOv3 into the system and comparison with the performance of TensorFlow Lite and Keras' real-time object detection should be considered. While the initial goal was to develop a mobile application for user convenience, enhancing the scalability of the CNN and training speed should take precedence.

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